Off on a Tangent

Meditations on the derivative

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Contents

1. Introduction **5**

1.A. Prerequisites . . . 5

1.B. Linear algebra notation . . . 5

1.B.1. Some important vector spaces . . . 5

1.B.2. Matrix notation . . . 5

1.C. Little-oh notation . . . 7

1.D. Norms of linear maps . . . 8

2. Single variable derivatives **11**

2.A. Slope of tangent lines . . . 11

2.B. Best linear approximation . . . 13

2.C. Rules of differentiation . . . 15

2.C.1. Sum and scalar multiples rule . . . 16

2.C.2. Product and quotient rules . . . 16

2.C.3. Chain rule . . . 17

2.C.4. Some rules without proofs . . . 24

2.D. Levels of differentiability . . . 25

2.D.1. C^n functions . . . 25

2.D.2. Smooth functions . . . 26

3. Multivariable derivatives **29**

3.A. Detailed example . . . 29

3.B. Definition of the derivative . . . 29

3.C. Computing derivatives . . . 30

3.C.1. Sum and scalar multiples rule . . . 30

3.C.2. Chain rule . . . 30

3.C.3. Differentiability by components . . . 31

3.C.4. Partial derivatives . . . 32

3.C.5. Jacobian matrix . . . 34

3.D. Levels of differentiability 34
3.D.1. Continuous differentiability 34
3.D.2. C^n and smooth functions 36
3.E. Inverse function theorem 36

4. Manifolds **37**
4.A. Defining Manifolds 37
4.A.1. Charts and atlases 37
4.A.2. Definition and examples 38
4.B. Smooth functions 40
4.B.1. Smooth real-valued functions 40
4.B.2. Smooth functions between manifolds 40
4.B.3. Diffeomorphisms 41

5. Tangent Spaces **43**
5.A. Tangent space 43
5.A.1. Curves based at a point 43
5.A.2. Vector space structure on the tangent space 43
5.A.3. Derivations 43
5.B. Pushing tangent vectors forward 43

1. Introduction

1.A. Prerequisites

This text assumes familiarity with metric space theory and with linear algebra.

1.B. Linear algebra notation

1.B.1. Some important vector spaces

Definition 1.1. We write $\mathbb{R}^n$ to denote the vector space of lists of n numbers. We will sometimes write its elements as horizontal lists with commas, as in $(a_1, \ldots, a_n)$, and sometimes as vertical column vectors, as in

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

Definition 1.2. We write $\mathcal{M}_{n,m}$ for the set of all $n \times m$ matrices (ie, the matrices with n rows and m columns). This is naturally a vector space.

Definition 1.3. If V and W are both vector spaces, we write $\mathcal{L}(V, W)$ for the set of all linear maps $V \to W$. This is naturally a vector space.

1.B.2. Matrix notation

Definition 1.4. Suppose V is a finite dimensional vector space and $B = (v_1, \ldots, v_n)$ is a basis for V. Then any $v \in V$ can be written uniquely as $a_1 v_1 + \cdots + a_n v_n$ for some scalars $a_1, \ldots, a_n \in \mathbb{R}$, and we define the *representation of v with respect to* B, denoted $[v]_B$, to be the column vector with n entries that records the scalars $a_1, \ldots, a_n$. In other words,

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

Lemma 1.5. *If* V *is an* n*-dimensional vector space and* $B = (v_1, \ldots, v_n)$ *is a basis for* V*, then the function* $V \to \mathbb{R}^n$ *given by* $v \mapsto [v]_B$ *is an isomorphism, with inverse given by*

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \mapsto a_1 v_1 + \cdots + a_n v_n.$$

Definition 1.6. Suppose $\ell : V \to W$ is a linear map between finite dimensional vector spaces. Let $B = v_1, \ldots, v_m$ be a basis for V and let C be a basis for W. The *matrix representation of* ℓ *with respect to* B *and* C, denoted $[\ell]_{B,C}$, is the $n \times m$ matrix whose ith column of $[\ell]_{B,C}$ is $[\ell(v_i)]_C$. In other words,

$$[\ell]_{B,C} = \begin{bmatrix} [\ell(v_1)]_C & \cdots & [\ell(v_m)]_C \end{bmatrix}.$$

Lemma 1.7. *If* $\ell : V \to W$ *is a linear map between finite dimensional vector spaces and* B *and* C *are bases for* V *and* W*, respectively, then*

$$[\ell]_{B,C}[v]_B = [\ell(v)]_C$$

for any $v \in V$*.*

Lemma 1.8. *If* $\ell : U \to V$ *and* $\ell' : V \to W$ *are linear maps between finite dimensional vector spaces, and* A, B *and* C *are bases for* U, V*, and* W*, respectively, then*

$$[\ell' \circ \ell]_{A,C} = [\ell']_{B,C}[\ell]_{A,B}.$$

Lemma 1.9. *If* $\ell : V \to W$ *is a linear map between finite dimensional vector spaces and* B *and* C *are bases for* V *and* W*, respectively, then the rank of the matrix* $[\ell]_{B,C}$ *equals the dimension of the range* $\ell(V)$*.*

Lemma 1.10. *Suppose that* V *and* W *are* m*- and* n*-dimensional vector spaces, respectively, and that and* B *and* C *are bases for* V *and* W*, respectively. Then the function* $\ell \mapsto [\ell]_{B,C}$ *is an isomorphism*

$$\mathcal{L}(V, W) \longrightarrow \mathcal{M}_{n,m}.$$

Definition 1.11. If $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map, the *standard matrix representation* of ℓ, denoted $[\ell]$, is the matrix of ℓ with respect to the standard bases on $\mathbb{R}^m$ and $\mathbb{R}^n$.

Lemma 1.12. *If* A *is a* $n \times m$ *matrix and* $\ell_A : \mathbb{R}^m \to \mathbb{R}^n$ *is the linear map* $\ell_A(v) = Av$*, then*

$$[\ell_A] = A.$$

1.C. Little-oh notation

Throughout this section, let X be a metric space and let x_0 be a point in X.

Definition 1.13. A function $g : X \setminus \{x_0\} \to \mathbb{R}$ is *positive* if $g(x) > 0$ for all $x \in X \setminus \{x_0\}$. Similarly, g is *non-negative* if $g(x) geq 0$ for all $x \in X \setminus \{x_0\}$.

Suppose g is a positive function. Definition 1.14 below formulates what it means for a non-negative function $f : X \setminus \{x_0\} \to \mathbb{R}$ to be "little-oh of g as x tends to x_0." Intuitively, this condition means that f is *much smaller* than g near x_0.

Definition 1.14. A function $f : X \setminus \{x_0\} \to \mathbb{R}$ is *little-oh of* g *as* x *tends to* x_0, written

$$f(x) = o(g(x)) \text{ as } x \to x_0$$

if, for every $\epsilon > 0$, there exists an open neighborhood U of x_0 such that $f(x) \leqslant \epsilon g(x)$ for all $x \in U \setminus \{x_0\}$. Equivalently, this means that

$$\lim_{x \to x_0} \frac{|f(x)|}{g(x)} = 0.$$

When x_0 can be inferred from context, we write simply $f = o(g)$.

Remark 1.15. It's worth noting that the use of the symbol "=" above is mathematically abusive. The left-hand side of the "=" is a function and the right-hand side is a property of functions; of course, a function cannot literally be equal to a property. Rather, the "=" is being used here to mean that the function on the left-hand side has the property described on the right-hand side. As annoying as it is, this abuse of notation is fairly standard, so it's probably best to just get used to it.

Exercise 1.16. Prove that the two conditions in definition 1.14 are in fact equivalent.

Exercise 1.17. Let $X = \mathbb{R}$ and $x_0 = 0$. For each of the functions f and g described below, sketch graphs of f and g and then determine whether or not $f = o(g)$ as $x \to 0$.

(a) $f(x) = x$ and $g(x) = x^2$.

(b) $f(x) = x$ and $g(x) = |x|$.

(c) $f(x) = x^2$ and $g(x) = |x|$.

Exercise 1.18. Let $X = \mathbb{R}^2$ and $x_0 = 0$. For each of the following functions f, determine whether or not $f(x) = o(|x|)$ as $x \to 0$.

(a) $f(x, y) = x - y$.

(b) $f(x, y) = x^2 + y^2$.

Exercise 1.19 ("Smaller than small is still small")**.** Suppose $f_1, f_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions such that $|f_1(x)| \leqslant |f_2(x)|$ for all x in a punctured neighborhood of x_0, and $f_2(x) = o(g(x))$ as $x \to x_0$. Then $f_1(x) = o(g(x))$ also.

Exercise 1.20. For a fixed positive function $g : X \setminus \{x_0\} \to \mathbb{R}$, prove that the set of functions $f : X \setminus \{x_0\} \to \mathbb{R}$ which are $o(g)$ as $x \to x_0$ is a vector space under the natural operations. In other words, verify the following three facts.

(1) The zero function is $o(g(x))$.

(2) ("Scalar multiples of small are still small") If $c \in \mathbb{R}$ is a scalar and $f(x) = o(g(x))$, then

$$cf(x) = o(g(x)).$$

(3) ("Sum of smalls is still small") If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$, then

$$(f_1 + f_2)(x) = o(g(x)).$$

Exercise 1.21. Suppose $f_1, f_2, g_1, g_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions with g_1, g_2 positive such that $f_1(x) = o(g_1(x))$ and $f_2(x) = o(g_2(x))$ as $x \to x_0$. Prove that $(f_1 \cdot f_2)(x) = o((g_1 \cdot g_2)(x))$.

1.D. Norms of linear maps

Definition 1.22. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Then we define the *operator norm*, or just *norm*, of ℓ, denoted $\|\ell\|$, by

$$\|\ell\| = \sup\{|\ell(v)| : |v| \leqslant 1\}.$$

Exercise 1.23. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that

$$\|\ell\| = \inf\{\lambda \in \mathbb{R} : |\ell(v)| \leqslant \lambda|v| \text{ for all } v \in \mathbb{R}^m\}.$$

In particular, $|\ell(v)| \leqslant \|\ell\| \cdot |v|$ for all $v \in \mathbb{R}^m$.

Exercise 1.24. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that $\ell = 0$ if and only if $\|\ell\| = 0$.

Exercise 1.25 (Subadditivity)**.** Suppose $\ell, \ell' : \mathbb{R}^m \to \mathbb{R}^n$ are both linear maps. Show that

$$\|\ell + \ell'\| \leqslant \|\ell\| + \|\ell'\|.$$

Definition 1.26. Let $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ denote the set of all linear maps $\mathbb{R}^m \to \mathbb{R}^n$. Define the distance between $\ell, \ell' \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ by $\|\ell - \ell'\|$. It follows from exercises 1.24 and 1.25 that this distance function makes $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ a metric space.

Exercise 1.27 ("Evaluation is continuous")**.** Suppose $v_0 \in \mathbb{R}^m$ is a fixed vector. Then the "evaluate at v_0" function $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathbb{R}^n$ given by $\phi \mapsto \phi(v_0)$ is continuous. *Possible hint*. Since the "evaluate at v_0" function is linear, it is sufficient to check continuity at 0.

Exercise 1.28 (Submultiplicativity)**.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ and $\ell' : \mathbb{R}^n \to \mathbb{R}^p$ are both linear maps. Show that

$$\|\ell' \circ \ell\| \leqslant \|\ell'\| \cdot \|\ell\|.$$

Give an example to show that this inequality can be strict.

Exercise 1.29 ("Composition is continuous")**.** (a) Suppose $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is a linear map. Show that the "postcompose with ℓ" function $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$ given by $\phi \mapsto \ell \circ \phi$ is continuous.

(b) Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that the "precompose with ℓ" function $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^p) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$ given by $\phi \mapsto \phi \circ \ell$ is continuous.

Possible hint (for both parts). Because both the "postcompose with ℓ" and the "precompose with ℓ" maps are linear, it is sufficient to show that the maps are continuous at 0. You will likely find exercise 1.28 useful.

Exercise 1.30 ("Linear functions of small functions are still small")**.** Suppose $r : \mathbb{R}^m \to \mathbb{R}^n$ is a function such that $|r(v)| = o(|v|)$ as $v \to 0$. If $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is linear, prove that $|\ell(r(v))| = o(|v|)$ also.

Lemma 1.31. *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a nonzero linear map. Then there exists $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$.*

Proof. The "unit disk"

$$D = \{x \in \mathbb{R}^m : |x| \leqslant 1\}$$

is a nonempty, connected, and compact subset of $\mathbb{R}^m$. The function $v \mapsto |\ell(v)|$ is also a continuous function $D \to \mathbb{R}$, so its image $|\ell(D)|$ must also be a nonempty, connected, and

compact subset of $\mathbb{R}$. In other words, it must be a closed interval in $\mathbb{R}$. This means that there exists $v_0 \in D$ such that

$$|\ell(v_0)| = \sup |\ell(D)| = \|\ell\|.$$

Since $\ell \neq 0$, we know $\|\ell\| \neq 0$ from exercise 1.24, which in turn means that $v_0 \neq 0$. To prove that $|v_0| = 1$, assume for a contradiction that $|v_0| < 1$, and let $c = 1/|v_0| > 1$. Then $|cv_0| = |c||v_0| = 1$, so $cv_0 \in D$. But then

$$|\ell(cv_0)| = |c\ell(v_0)| = c|\ell(v_0)| > |\ell(v_0)| = \sup |\ell(D)|,$$

which is absurd. □

Exercise 1.32. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map and $|\ell(v)| = o(|v|)$ as $v \to 0$. Prove that $\ell = 0$.

2. Single variable derivatives

The goal in this chapter is to formalize derivatives of functions whose input is a single real number and whose output is also a single real number.

2.A. Slope of tangent lines

Let S be a subset of $\mathbb{R}$ and $a \in S$ an interior point. We would like to define the tangent line to the graph of a function $f : S \to \mathbb{R}$ at a. We know intuitively what "tangent line" should mean: a line that "just barely touches" the graph of f at the point $(a, f(a))$. But it's not immediately clear how to formalize this intuition.

One approach to formalization is to notice that our intuitive idea of "tangent line" can be approximated by secant lines, and secant lines can be formalized without ambiguity. More precisely, for small values of h, the point $a + h$ is close to a and still in S since a is an interior point of S. The slope of the secant line passing through the two points $(a, f(a))$ and $(a + h, f(a + h))$ is computed by the quantity

$$\frac{f(a+h) - f(a)}{h}.$$

This quantity is often called a *difference quotient*. See figure 2.1 for a geometric explanation of this difference quotient. By letting $h \to 0$, we get closer and closer to what we intuitively understand to be the slope of the tangent line.

Definition 2.2. A function $f : S \to \mathbb{R}$ is *differentiable* at an interior point $a \in S$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. If this limit does exist, its value is called the *derivative of* f *at* a, and is denoted $f'(a)$.

Exercise 2.3. Use the definition of the derivative to determine whether or not each of the following functions f is differentiable at the given point a. If it is differentiable, calculate $f'(a)$. Check your answers by making sure they agree with geometric intuition and/or rules you learned during your introductory calculus course.

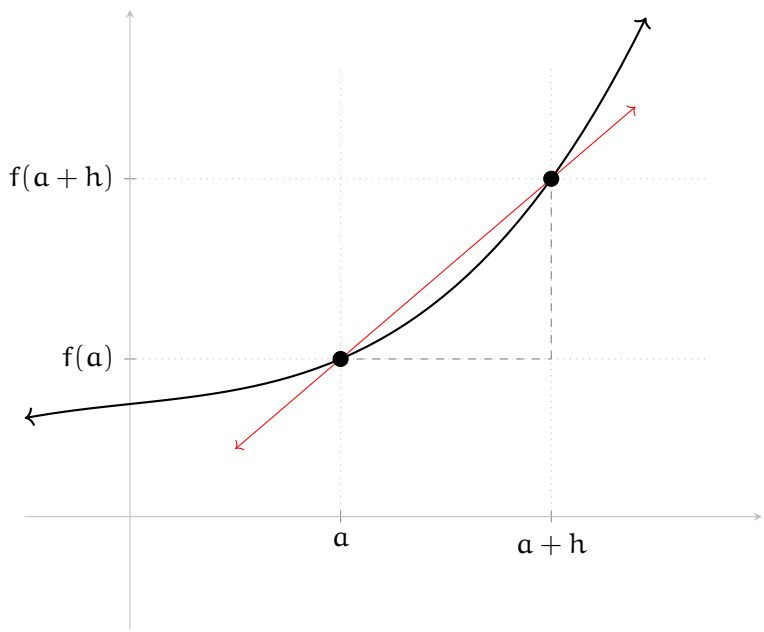

Figure 2.1.: The graph of a function f is depicted in black. The secant line passing through the two points $(a, f(a))$ and $(a+h, f(a+h))$ is depicted in red. The "run" between these points, is h. The "rise" between these points is $f(a+h) - f(a)$. Thus the slope of the secant line, computed as "rise over run," is precisely the difference quotient $(f(a+h) - f(a))/h$.

(a) $f(x) = |x|$ at $a = 0$.

(b) $f(x) = x$ at $a = 0$.

(c) $f(x) = x^2$ at $a = 0$.

(d) $f(x) = x^2$ at $a = 2$.

(e) $f(x) = \begin{cases} 1 & x \geqslant 0 \\ 0 & x < 0 \end{cases}$ at $a = 0$.

Exercise 2.4 (Power rule for positive integer exponents)**.** Prove that, if $f(x) = x^n$ for a positive integer n, then $f'(a) = na^{n-1}$. *Hint.* Recall the binomial theorem, which says that

$$(a+h)^n = \sum_{k=0}^{n} \binom{n}{k} a^k h^{n-k}.$$

Exercise 2.5 (Differentiability implies continuity)**.** Prove that if $f : U \to \mathbb{R}$ is differentiable at $a \in U$, then f is also continuous at a.

2.B. Best linear approximation

Before proceeding further, we reinterpret the definition of the derivative in a way that will be better suited to generalization to the multivariable setting. Suppose a function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. The tangent line to the graph at the point $(a, f(a))$ is the graph of the function

$$t(x) = f'(a)(x - a) + f(a).$$

Since this function t is "close" to f near a, we should have that

$$f(a+h) - f(a) \approx t(a+h) - t(a) = f'(a)h$$

for small values of h. See figure 2.6.

Notice that the function $h \mapsto f'(a)h$ is a linear map $\mathbb{R} \to \mathbb{R}$ in the sense of linear algebra. One way to interpret the following result is that the function $h \mapsto f'(a)h$ is the *only* linear map that is a good approximation to $h \mapsto f(a+h) - f(a)$.

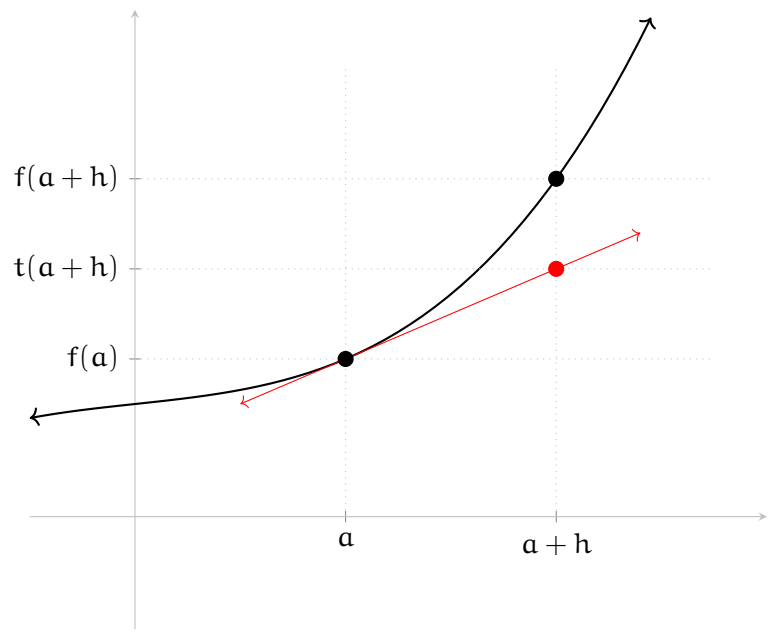

Figure 2.6.: The graph of a function f is depicted in black, and its tangent line t at a is depicted in red. Then the vertical distance $f(a+h) - f(a)$ is approximated by the vertical distance $t(a+h) - t(a)$. Since the tangent line has slope $f'(a)$, this "rise" can be computed as $f'(a)$ times the "run" (in other words, $f'(a)h$). As h gets smaller, this approximation gets better.

Proposition 2.7. *A function* $f : S \to \mathbb{R}$ *is differentiable at an interior point* $a \in S$ *if and only if there exists a linear map* $\ell : \mathbb{R} \to \mathbb{R}$ *such that*

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

Moreover, if there exists such a linear map ℓ*, then* ℓ *is uniquely determined: it must be given by*

$$\ell(h) = f'(a)h$$

for all $h \in \mathbb{R}$*.*

Proof. Let ℓ be a linear map $\ell(h) = ch$, where c is some scalar. Observe that

$$\begin{aligned}\frac{|f(a+h) - f(a) - \ell(h)|}{|h|} &= \left|\frac{f(a+h) - f(a) - ch}{h}\right| \\ &= \left|\frac{f(a+h) - f(a)}{h} - c\right|.\end{aligned}$$

This leads to the following sequence of "if and only if" statements.

$$\begin{aligned}|f(a+h) - f(a) - \ell(h)| = o(|h|) &\iff \lim_{h\to 0} \frac{|f(a+h) - f(a) - \ell(h)|}{|h|} = 0 \\ &\iff \lim_{h\to 0} \left|\frac{f(a+h) - f(a)}{h} - c\right| = 0 \\ &\iff \lim_{h\to 0} \left(\frac{f(a+h) - f(a)}{h} - c\right) = 0 \\ &\iff \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = c\end{aligned}$$

The statement of the proposition follows. □

Definition 2.8. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, we define $df_a : \mathbb{R} \to \mathbb{R}$ to be the linear map $df_a(h) = f'(a)h$.

In other words, df_a is the only good linear approximation to the function $h \mapsto f(a+h) - f(a)$. Being the only good linear approximation, it is also the best linear approximation, whence the name of this section.

2.C. Rules of differentiation

The goal in this section is to prove the rules for differentiation that one encounters in an introductory calculus course.

2.C.1. Sum and scalar multiples rule

Both of the following can be proved using either the definition of the derivative 2.2 or the equivalent "best linear approximation" characterization of proposition 2.7. My suggestion is that you try both methods for both of these exercises. When you use proposition 2.7, you may find exercise 1.20 useful.

Exercise 2.9 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, then $f + g$ is also differentiable at a and

$$(f + g)'(a) = f'(a) + g'(a).$$

Exercise 2.10 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$(cf)'(a) = cf'(a).$$

2.C.2. Product and quotient rules

Proposition 2.11 (Product rule)**.** *Suppose* $f, g : U \to \mathbb{R}$ *are both differentiable at* $a \in U$*. Then* fg *is also differentiable at* a *and*

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proof. The proof is a clever algebraic manipulation of difference quotients. The key trick is introducing a "cross term" of the form $f(a)g(a + h)$, and then cancelling it out by also introducing its negative. More precisely, for any sufficiently small real number h, we have

$$\begin{aligned}
\frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h) + f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h) - f(a)}{h} \cdot g(a+h) + f(a) \cdot \frac{g(a+h) - g(a)}{h}.
\end{aligned}$$

Taking the limit as $h \to 0$ yields the result. □

Exercise 2.12. Check your understanding of the above proof of the product rule.

(a) Could you have instead introduced a cross term of the form $f(a + h)g(a)$?

(b) At what point is exercise 2.5 tacitly used in the above proof?

Exercise 2.13. Reprove the scalar multiples rule (exercise 2.10) using the product rule.

Exercise 2.14 (Quotient rule)**.** Suppose $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, and that $g(a) \neq 0$. Prove that f/g is also differentiable at a and

$$(f/g)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Hint. Observe that

$$\frac{(f/g)(a+h) - (f/g)(a)}{h} = \frac{\frac{f(a+h)}{g(a+h)} - \frac{f(a)}{g(a)}}{h}.$$

Multiply both the numerator and denominator by $g(a+h)g(a)$ to clear the denominators inside the numerator, and then introduce the cross term $f(a)g(a)$. Be sure to notice at what point you use exercise 2.5.

Exercise 2.15 (Power rule for integer exponents)**.** Extend the result of exercise 2.4 to arbitrary integer exponents. *Hint.* Use the quotient rule for negative integers. The exponent zero is a special case; deal with it separately.

2.C.3. Chain rule

Proposition 2.16 (Chain rule)**.** *Suppose that S and T are both subsets of $\mathbb{R}$, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}$ is also differentiable at a, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Let us discuss two different proofs of this important result.

2.C.3.A. First proof

The first proof uses the definition of the derivative 2.2, but there is a subtle technicality involved; so, let us discuss it informally first before diving into the formal proof.

The idea involved is again a clever rewriting of difference quotients. Again, we introduce a cross term, which in this is $f(a+h) - f(a)$. But this time we introduce this cross term "multiplicatively" rather than "additively." More precsiely, we have

$$\begin{aligned}\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} &= \frac{g(f(a+h)) - g(f(a))}{h} \\ &= \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \cdot \frac{f(a+h) - f(a)}{h}.\end{aligned}$$

The difference quotient on the right is precisely $f'(a)$. Thus, it would be sufficient to prove

$$\lim_{h\to 0} \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = g'(f(a)). \tag{2.17}$$

This formula looks an awful lot like the difference quotient that is used to define the derivative of g at $f(a)$. In fact, since f is continuous at a by exercise 2.5, we have $f(a+h) \approx f(a) + h$ for small values of h, which means that

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \approx \frac{g(f(a)+h) - g(f(a))}{f(a)+h-f(a)} = \frac{(g\circ f)(a+h) - (g\circ f)(a)}{h}$$

for small values of h. Taking the limit as $h \to 0$ on the right-hand side yields precisely $g'(f(a))$, so equation (2.17) seems like a reasonable conjecture.

The problem is that equation (2.17) isn't true. Issues arise when $f(a+h) = f(a)$ for arbitrarily small values of h, which leads to an infinite sequence of divisions by 0.

Example 2.18. Let

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let $g(x) = x^2$. Observe that $g'(f(0)) = 0$. You will show later (in exercise 2.29) that f is in fact differentiable at 0, so these functions do satisfy the hypothesis of the chain rule. However, the limit as $h \to 0$ of

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = \frac{(h^2 \sin(1/h))^2}{h^2 \sin(1/h)}$$

does not exist. For example, the sequence $h_k = 1/k\pi$ converges to 0 as $k \to \infty$, but the expression above is undefined along this sequence.

We will get around the kind of issue that is described by the above example by carefully studying the expression

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)}. \tag{2.19}$$

Observe that, when $f(a+h) \neq f(a)$, then this expression is precisely the difference quotient that calculates the slope of the secant line passing through $(f(a), g(f(a))$ and $(f(a+h), g(f(a+h))$. When $f(a+h) = f(a)$, this secant line doesn't make sense; but what does make sense is the tangent line! So we'll remove the undefinedness of expression (2.19) by extending it so that it takes the value $g'(f(a))$ whenever $f(a+h) = f(a)$.

This leads us to our first proof of the chain rule.

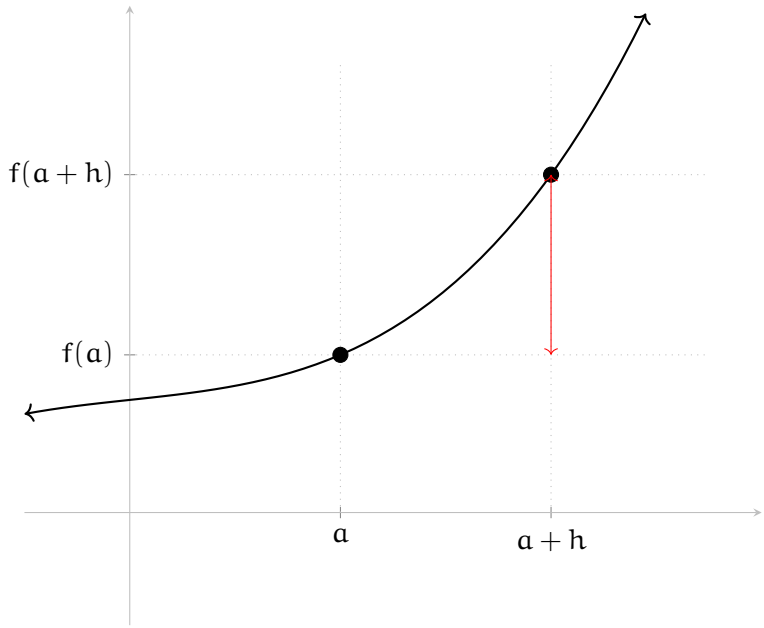

Figure 2.20.: The graph of f is depicted in black. The value of the function t on a small nonzero input value h is the vertical distance depicted in red.

First proof of the chain rule. Consider the function ϵ defined for small values of h by $\epsilon(h) = f(a+h) - f(a)$. See figure 2.20. You will verify in exercise 3.10 below that ϵ is continuous at 0.

Next, define a function σ by the following.

$$\sigma(k) = \begin{cases} \dfrac{g(f(a)+k) - g(f(a))}{k} & \text{if } k \neq 0 \\ g'(f(a)) & \text{if } k = 0 \end{cases}$$

In other words, σ is the function that, on some small nonzero input k, outputs the slope of the secant line connecting the two points on the graph of g where $x = f(a)$ and $x = f(a) + k$. See figure 2.21. It follows from the definition of the derivative of g at $f(a)$ that the function σ is continuous at 0.

Observe that

$$\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} = \sigma(\epsilon(h)) \cdot \frac{f(a+h) - f(a)}{h} \tag{2.22}$$

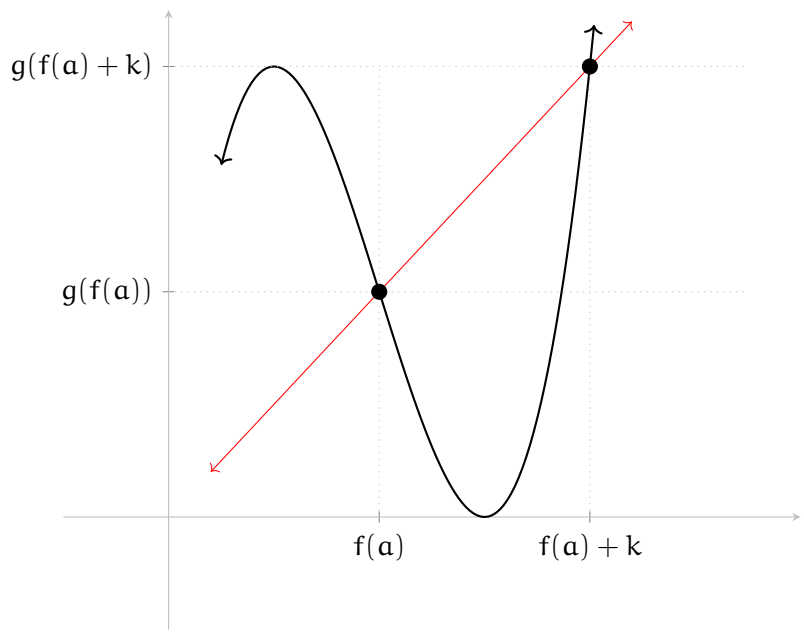

Figure 2.21.: The graph of g is depicted in black. The output of the function σ on a small nonzero input value k is the slope of the red secant line passing through the two points $(f(a), g(f(a)))$ and $(f(a)+k, g(f(a)+k))$.

for all small values of h, as you will verify in exercise 3.10 below. Thus

$$\begin{aligned}\lim_{h\to 0}\frac{(g\circ f)(a+h)-(g\circ f)(a)}{h} &= \lim_{h\to 0}\sigma(\epsilon(h))\cdot\frac{f(a+h)-f(a)}{h}\\ &\overset{(*)}{=} \sigma(\epsilon(0))f'(a)\\ &= g'(f(a))f'(a)\end{aligned}$$

where we have used the continuity of ϵ and σ and the definition of the derivative of f at a for the step labeled $(*)$ above. This proves the chain rule. □

Exercise 2.23. Verify the missing details in the above proof. Specifically, check the following.

(a) Prove that ϵ is continuous at 0.

(b) Prove equation (2.22). *Hint.* As suggested by our analysis before the proof of the chain rule, split up your proof of equation (2.22) into two cases: when $f(a+h)\neq f(a)$, and when $f(a+h)=f(a)$.

2.C.3.B. Second proof

Our second proof of the chain rule will use the "best linear approximation" interpretation of derivatives provided by proposition 2.7.

Second proof of the chain rule. By proposition 2.7, it is sufficient to prove that

$$g(f(a+h))-g(f(a))-dg_{f(a)}(df_a(h))=o(|h|) \text{ as } h\to 0.$$

Define the following "error in approximation" functions.

$$\begin{aligned}r(h) &= f(a+h)-f(a)-df_a(h)\\ s(k) &= g(f(a)+k)-g(f(a))-dg_{f(a)}(k)\end{aligned}$$

See figure 2.24.

Since

$$f(a+h)=f(a)+df_a(h)+r(h)$$

by definition of r, we have

$$\begin{aligned}g(f(a+h)) &= g(f(a)+df_a(h)+r(h))\\ &= g(f(a))+dg_{f(a)}(df_a(h)+r(h))+s(df_a(h)+r(h))\\ &= g(f(a))+dg_{f(a)}(df_a(h))+dg_{f(a)}(r(h))+s(df_a(h)+r(h))\end{aligned}$$

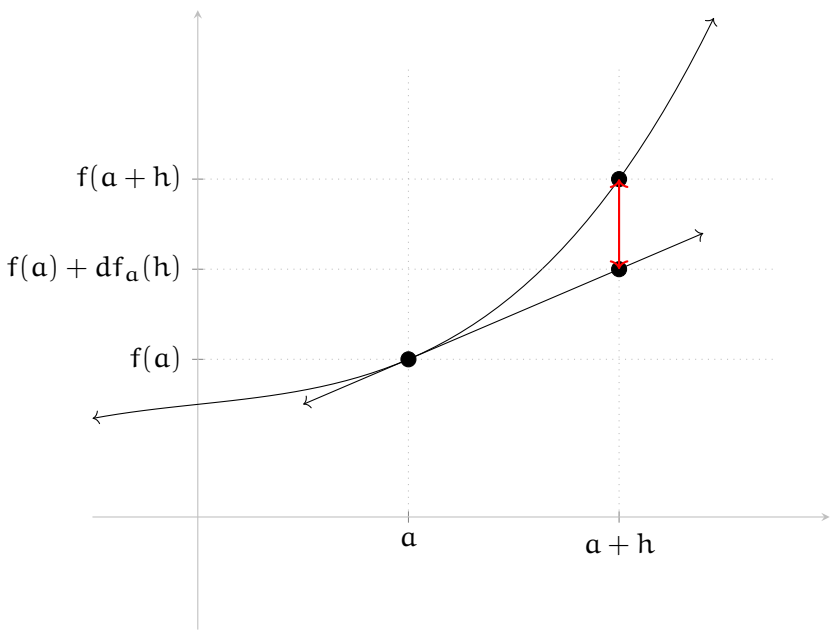

Figure 2.24.: The graphs of a function f and its tangent line $f(a) + df_a$ are depicted in red. On a small input h, the "error in approximation" function r outputs the vertical distance indicated in red.

where we use the definition of s on input $k = df_a(h) + r(h)$ for the second step, and the fact that $dg_{f(a)}$ is linear for the third. Thus

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)).$$

We therefore want to prove that the right-hand side is $o(|h|)$.

Since

$$dg_{f(a)}(r(h)) = g'(f(a)) \cdot r(h)$$

is just a scalar multiple of r, and $r = o(|h|)$, we see that $dg_{f(a)}(r) = o(|h|)$ also. So it is sufficient to prove that

$$s(df_a(h) + r(h)) = o(|h|).$$

We've now hit the technical part, so let's hit pause.

To see why this is technical, and for essentially the same reasons as the previous proof, notice that we're trying to prove that

$$\lim_{h \to 0} \frac{|s(df_a(h) + r(h)|)}{|h|} = 0.$$

To prove this, it's tempting to introduce a multiplicative cross-term as follows.

$$\frac{|s(df_a(h) + r(h)|)}{|h|} = \frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|} \cdot \frac{|df_a(h) + r(h)|}{|h|}$$

But notice that $df_a(h) + r(h) = f(a+h) - f(a)$ by definition of r, so this cross term is problematic for the same reasons as before! If $f(a+h) = f(a)$ for h arbitrary close to 0, we have an infinite sequence of divisions by zeroes that causes problems.

We'll deal with this problem in essentially the same way as before: namely, we extend the definition of

$$\frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|}$$

so that it is also defined when $f(a+h) = f(a)$.

Second proof of the chain rule, continued. Define $\eta(k) = s(k)/k$. Observe that $\eta(0) = 0$. Moreover, since $s(k) = o(|k|)$ by proposition 2.7, we see that η is continuous at 0. Then

$$\begin{aligned}\frac{|s(df_a(h) + r(h))|}{|h|} &= \frac{|\eta(df_a(h) + r(h))| \cdot |df_a(h) + r(h)|}{|h|} \\ &\leqslant |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|}\right).\end{aligned}$$

Observe that $df_a + r$ is continuous at 0 by **??** below; moreover, we have just seen that η is continuous at $(df_a + r)(0) = 0$. Thus the composite $\eta \circ (df_a + r)$ is also continuous at 0. Thus

$$\lim_{h \to 0} |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|} \right) = 0,$$

where we have used the fact that $|r(h)| = o(|h|)$ to see that the parenthetical expression has a finite limit (namely, $|f'(a)|$). So, by the squeeze theorem, we can therefore conclude that $|s(df_a(h) + r(h))| = o(|h|)$. □

Exercise 2.25. Check that the function r is continuous at 0.

Notice that in the first proof, the trick to overcoming the technicality was defining the function σ at $k = 0$. In the second proof, the trick to overcoming the same technicality was defining η at $k = 0$. These two tricks are really the same trick, as shown by the following.

Exercise 2.26. Prove that the function η from the second proof and the function σ from the first proof are related by the equation

$$\sigma(k) = g'(f(a)) + \eta(k).$$

2.C.3.C. Applications of the chain rule

Exercise 2.27 (Derivatives of inverses)**.** Suppose S and T are both subsets of $\mathbb{R}$ and $f : S \to T$ is bijective, with inverse function $f^{-1} : T \to S$. Suppose further that f is differentiable at an interior point $a \in S$, that $f'(a) \neq 0$, and that $b = f(a)$ is an interior point of T. Prove that f^{-1} is differentiable at b and

$$(f^{-1})'(b) = \frac{1}{f'(a)}.$$

Exercise 2.28 (Power rule for fractional exponents)**.** Extend the result of exercise 2.15 to arbitrary rational exponents (ie, to exponents that can be written in the form m/n where m and n are both integers).

2.C.4. Some rules without proofs

Here are some other rules you probably remember from calculus. We won't prove them here, but you should feel free to use them.

(1) If $f(x) = e^x$, then $f'(x) = e^x$.

(2) If $f(x) = \ln(x)$, then $f'(x) = 1/x$.

(3) If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.

(4) If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.

(5) If $f(x) = x^r$ for some *real* number r, then $f'(x) = rx^{r-1}$.

Note that by doing exercise 2.28, you have already proved the most common cases of the power rule (5).

The proofs of these facts are omitted not because the proofs are particularly difficult, but because we would have to go on a long detour to even define the functions involved sufficiently rigorously to prove anything about them. That would distract us from our object of meditation.

Exercise 2.29. (a) Prove that the function

$$f_1(x) = \begin{cases} x\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous at 0 but is not differentiable at 0. In particular words, $f \in C^0(\mathbb{R}) \setminus C^1(\mathbb{R})$.

(b) Prove that the function

$$f_2(x) = \begin{cases} x^2\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable at 0 but the derivative f_2' is not continuous at 0.

2.D. Levels of differentiability

2.D.1. C^n functions

Definition 2.30. Suppose U is an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ is a function. Then f is *differentiable* if f is differentiable at every $a \in U$. If the function $f' : U \to \mathbb{R}$ is also continuous, then f is *continuously differentiable*.

Definition 2.31. Let U be an open subset of $\mathbb{R}$. For any non-negative integer n, we define the set $C^n(U)$ inductively as follows.

- $C^0(U)$ is the set of continuous functions $f : U \to \mathbb{R}$.

- $C^1(U)$ is the set of functions f such that f' exists and is in $C^0(U)$. Concretely, this means that f is continuously differentiable.
- $C^2(U)$ is the set of functions $f : U \to \mathbb{R}$ such that the derivative f' exists and is in $C^1(U)$. Concretely, this means that the the derivative of the derivative $(f')'$, called the *second derivative* of f and written f'' or $f^{(2)}$, exists and is continuous on U.
- In general, $C^{n+1}(U)$ is the set of functions $f : U \to \mathbb{R}$ such that f' exists and is in $C^n(U)$. This means that the $(n+1)$*st derivative* $f^{(n+1)} = (f^{(n)})'$ exists and is continuous on U.

Exercise 2.32. Prove that, for every non-negative integer n, $C^n(U)$ is a subspace of the vector space of all functions $U \to \mathbb{R}$.

Exercise 2.33. Prove that, if $f, g \in C^n(U)$, then the product fg is also in $C^n(U)$.

We have an infinite chain of inclusions

$$\{\text{all functions } U \to \mathbb{R}\} \supseteq C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq C^3(U) \supseteq \cdots .$$

The following exercise shows that all of these inclusions are strict; in other words $C^n(U) \supseteq C^{n+1}(U)$ for all n.

Exercise 2.34. For every non-negative integer n, let

$$f_n(x) = x^n|x|.$$

Prove that f_n is in $C^n(\mathbb{R})$ but not $C^{n+1}(\mathbb{R})$. *Hint.* Prove that $f'_{n+1} = (n+2)f_n$. Then use induction.

2.D.2. Smooth functions

Definition 2.35. A function $f : U \to \mathbb{R}$ is *infinitely differentiable* or *smooth* if it is in $C^n(U)$ for all n. The set of smooth functions is denoted $C^\infty(U)$ or $\mathcal{O}(U)$.

Exercise 2.36. Consider the function $\psi : \mathbb{R} \to \mathbb{R}$ defined as follows.

$$\psi(x) = \begin{cases} e^{1/(x^2-1)} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

See figure 2.37 for its graph. Prove that ψ is smooth. *Hint.* First prove directly from definitions that ψ is continuously differentiable. Then use induction to prove that it is in $C^n(\mathbb{R})$ for all n.

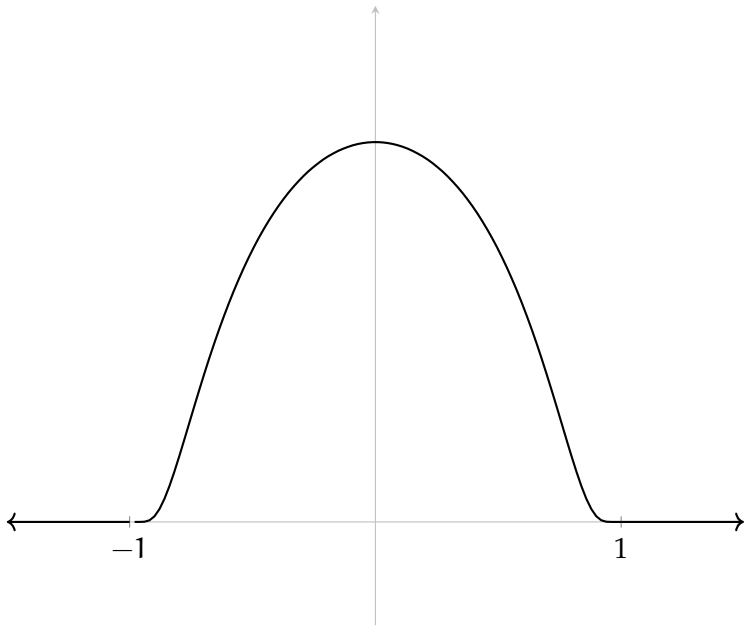

Figure 2.37.: The graph of the function ψ from exercise 2.36.

Exercise 2.38. Let $K = [a, b]$ be a compact interval in $\mathbb{R}$ and let U be an open subset containing K. Prove that there exists a smooth function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$. *Hint.* Since U is open, show that there exist $a' < a$ and $b' > b$ in U. Thus, if we let $I = [a', b']$, then $K \subsetneq I \subseteq U$. The characteristic function

$$\chi_I(x) = \begin{cases} 1 & \text{if } x \in I \\ 0 & \text{if } x \notin I \end{cases}$$

is not smooth, since it's not even continuous; obtain f by modifying χ_I on the intervals $[a', a]$ and $[b, b']$ using inspiration from the function ψ of exercise 2.36.

3. Multivariable derivatives

3.A. Detailed example

3.B. Definition of the derivative

Throughout, S will denote a subset of $\mathbb{R}^m$.

Definition 3.1. A function $f : S \to \mathbb{R}^n$ is *differentiable at* an interior point $a \in S$ if there exists a linear map $\ell : \mathbb{R}^m \to \mathbb{R}^n$ such that

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \text{ as } v \to 0.$$

It turns out that there can exist at most one linear map that has the above property.

Lemma 3.2. *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. If ℓ and ℓ' are both linear maps $\mathbb{R}^m \to \mathbb{R}^n$ such that*

$$|f(a+v) - f(a) - \ell(v)| = o(|v|) \textit{ and } |f(a+v) - f(a) - \ell'(v)| = o(|v|) \textit{ as } v \to 0,$$

then $\ell = \ell'$.

Proof. Let $\phi = \ell' - \ell$. Then ϕ is also a linear map $\mathbb{R}^m \to \mathbb{R}^n$. Moreover, observe that

$$\begin{aligned}
|\phi(v)| &= |\ell'(v) - \ell(v)| \\
&= |(f(a+v) - f(a) - \ell(v)) - (f(a+v) - f(a) - \ell'(v)| \\
&\leqslant |f(a+h) - f(a) - \ell(v)| + |f(a+v) - f(a) - \ell'(v)|.
\end{aligned}$$

Since both $|f(a+v) - f(a) - \ell(v)|$ and $|f(a+v) - f(a) - \ell'(v)|$ are $o(|v|)$, exercises 1.19 and 1.20 imply that $|\phi(v)| = o(|v|)$ also. Then exercise 1.32 implies that $\phi = 0$. □

Definition 3.3. Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. Then the *derivative* of f at a, denoted df_a, is the unique linear map $\mathbb{R}^m \to \mathbb{R}^n$ satisfying

$$|f(a+v) - f(a) - df_a(v)| = o(|v|) \text{ as } v \to 0.$$

This definition is very difficult to compute examples from. Virtually the only concrete example that we can calculate without much effort directly from the definition is the following.

Exercise 3.4. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Prove that ℓ is differentiable at every $a \in \mathbb{R}^m$ and $d\ell_a = \ell$.

Exercise 3.5. If $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, show that f must be continuous at a.

3.C. Computing derivatives

3.C.1. Sum and scalar multiples rule

Exercise 3.6 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}^n$ are both differentiable at an interior point $a \in S$, then $f + g$ is also differentiable at a and

$$d(f+g)_a = df_a + dg_a.$$

Exercise 3.7 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$d(cf)_a = c \cdot df_a.$$

3.C.2. Chain rule

The proof of the following is essentially the same as the second proof of the single variable chain rule given in section 2.C.3.B. Since this proof is basically a repeat, some details are omitted; you are asked to fill in the details in exercise 3.10 below.

Proposition 3.8 (Chain rule)**.** *Suppose that S and T are subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}^p$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}^p$ is also differentiable at a, and*

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

Proof. Define the "error in approximation" functions

$$r(v) = f(a+v) - f(a) - df_a(v)$$
$$s(w) = g(f(a)+w) - g(f(a)) - dg_{f(a)}(w)$$

and then observe that

$$g(f(a+v)) - g(f(a)) - dg_{f(a)}(df_a(v)) = dg_{f(a)}(r(v)) + s(df_a(v) + r(v)). \tag{3.9}$$

This means that

$$|g(f(a+v)) - g(f(a)) - dg_{f(a)}(df_a(v))| \leqslant |dg_{f(a)}(r(v))| + |s(df_a(h) + r(v))|.$$

Since $dg_{f(a)}$ is linear and $|r(v)| = o(|v|)$, exercise 1.30 guarantees that $|dg_{f(a)}(r(v))| = o(|h|)$ also. Thus it is sufficient to prove that $|s(df_a(v) + r(v))| = o(|v|)$. To do this, define $\eta(w) = |s(w)|/|w|$ and then notice that

$$\begin{aligned}\frac{|s(df_a(v) + r(v))|}{|v|} &= \eta(df_a(v) + r(v)) \cdot \frac{|df_a(v) + r(v)|}{|v|} \\ &\leqslant \eta(df_a(v) + r(v)) \left(\|df_a\| + \frac{|r(v)|}{|v|} \right).\end{aligned}$$

We now take the limit as $h \to 0$. Using the facts that $\eta \circ (df_a + r)$ is continuous at 0 and that $|r(v)| = o(|v|)$, we obtain the result. □

Exercise 3.10. (a) Prove equation (3.9).

(b) If you haven't done it already, do exercise 1.30.

(c) Prove that $\eta \circ (df_a + r)$ is continuous at 0.

3.C.3. Differentiability by components

Here we show that it's "enough" to consider the case $n = 1$.

Definition 3.11. For $j = 1, \ldots, n$, let $\pi_j : \mathbb{R}^n \to \mathbb{R}$ denote the jth *projection map*

$$\pi_j(x_1, \ldots, x_n) = x_j.$$

Then, for any function $f : S \to \mathbb{R}^n$, we define the jth *component function* $f_j : S \to \mathbb{R}$ to be the composite $\pi_j \circ f$.

Lemma 3.12. *A function $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$ if and only if the component function $f_j : S \to \mathbb{R}$ is differentiable at a for all $j = 1, \ldots n$. Moreover,*

$$df_a(v) = \begin{bmatrix} df_{1,a}(v) \\ \vdots \\ df_{n,a}(v) \end{bmatrix}. \tag{3.13}$$

Proof. If f is differentiable at a, then the composite $f_j = \pi_j \circ f$ is also differentiable at a by the chain rule 3.8. Moreover, since π_j is linear, we know that $d\pi_{j,f(a)} = \pi_j$ by exercise 3.4. Thus, by the chain rule,

$$df_{j,a}(v) = (d\pi_{j,f(a)} \circ df_a)(v) = \pi_j(df_a(v))$$

for all j, which is precisely equation (3.13). For the converse, we turn equation (3.13) on its head. Suppose f_j is differentiable at a for all j, and let $\ell : \mathbb{R}^m \to \mathbb{R}^n$ be the linear map

$$\ell(v) = \begin{bmatrix} df_{1,a}(v) \\ \vdots \\ df_{n,a}(v) \end{bmatrix}.$$

We now want to show that $|f(a+v) - f(a) - \ell(v)| = o(|v|)$ as $v \to 0$. Observe that the component functions of $v \mapsto f(a+v) - f(a) - \ell(v)$ are given by

$$v \mapsto \pi_j\left(f(a+v) - f(a) - \ell(v)\right) = f_j(a+v) - f_j(a) - df_{j,a}(v),$$

and we know that $|f_j(a+v) - f_j(a) - df_{j,a}(v)| = o(|v|)$. By exercise 3.14 below applied with the function $r(v) = f(a+v) - f(a) - \ell(v)$, we conclude that $|f(a+v) - f(a) - \ell(h)| = o(|v|)$. $\square$

Exercise 3.14. (a) For any vector $w \in \mathbb{R}^n$, show that

$$|w| \leqslant \sqrt{n} \max_j |\pi_j(w)|.$$

(b) Show that, if S is a neighborhood of 0 in $\mathbb{R}^m$ and $r : S \to \mathbb{R}^n$ is a function such that $|r_j(v)| = o(|v|)$ as $v \to 0$ for all $j = 1, \ldots, n$, then $|r(v)| = o(|v|)$ also. *Hint.* Use part (a).

3.C.4. Partial derivatives

Here we focus on the case $n = 1$. In other words, we consider functions whose input is m real numbers, and whose output is just a single number.

Definition 3.15. Suppose $f : S \to \mathbb{R}$ is a function and $a \in S$ is an interior point. The ith *partial derivative* of f at a, denoted $\partial_i f(a)$, is the limit

$$\lim_{h \to 0} \frac{f(a + he_i) - f(a)}{h}.$$

In other words, consider the function $\xi_i : \mathbb{R} \to \mathbb{R}^m$ given by $\xi_i(h) = a + he_i$. Then 0 is an interior point of $T = \xi_i^{-1}(S) \subseteq \mathbb{R}$, and $\partial_i f(a)$ is simply the derivative of the single variable function $f \circ \xi_i : T \to \mathbb{R}$ at 0.

Lemma 3.16. *If* $f : S \to \mathbb{R}$ *is differentiable at an interior point* $a \in S$*, then the* i*th partial derivative* $\partial_i f(a)$ *exists for all* $i = 1, \dots m$*, and the standard matrix representation* $[df_a]$ *of the linear map* $df_a : \mathbb{R}^m \to \mathbb{R}$ *is a* $1 \times m$ *matrix whose* i*th entry is* $\partial_i f(a)$*.*

$$[df_a] = \begin{bmatrix} \partial_1 f(a) & \partial_2 f(a) & \cdots & \partial_m f(a) \end{bmatrix}$$

Proof. Let r be the "error in approximation" function

$$r(v) = f(a + v) - f(a) - df_a(v),$$

so that $|r(v)| = o(|v|)$ as $h \to 0$. Then

$$\begin{aligned} 0 &= \lim_{h \to 0} \frac{|r(he_i)|}{|he_i|} \\ &= \lim_{h \to 0} \frac{|f(a + he_i) - f(a) - df_a(he_i)|}{|h|} \\ &= \lim_{h \to 0} \left| \frac{f(a + he_i) - f(a)}{h} - df_a(e_i) \right| \end{aligned}$$

which means that $df_a(e_i) = \partial f_a(e_i)$. □

Remark 3.17. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the vector

$$[df_a]^T = \begin{bmatrix} \partial_1 f(a) \\ \partial_2 f(a) \\ \vdots \\ \partial_m f(a) \end{bmatrix} \in \mathbb{R}^m$$

is often called the *gradient* of f at a, denoted $\nabla f(a)$.

Unfortunately, the converse to lemma 3.16 is *not* true, as shown by exercise 3.18. However, this behavior is fairly pathological; theorem 3.26 below will provide a converse under a fairly mild additional hypothesis.

Exercise 3.18. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as follows.

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq 0 \\ 0 & \text{if } (x_1, x_2) = 0 \end{cases}$$

Show that $\partial_1 f(0)$ and $\partial_2 f(0)$ both exist, but that f is not even continuous at 0, so cannot be differentiable by exercise 3.5. *Hint.* Approach the origin along the line $x_1 = x_2$.

3.C.5. Jacobian matrix

Definition 3.19. Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The *Jacobian matrix* at a, denoted $J(a)$, is the $n \times m$ matrix whose (j, i)-entry is $\partial_i f_j(a)$. In other words,

$$J(a) = \begin{bmatrix} \partial_1 f_1(a) & \partial_2 f_1(a) & \cdots & \partial_m f_1(a) \\ \partial_1 f_2(a) & \partial_2 f_2(a) & \cdots & \partial_m f_2(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_1 f_n(a) & \partial_2 f_n(a) & \cdots & \partial_m f_n(a) \end{bmatrix}$$

Note that lemma 3.16 tells us that all of these partial derivatives exist.

Theorem 3.20. *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. Then*

$$[df_a] = J(a).$$

Exercise 3.21. Prove theorem 3.20. *Hint.* Use lemmas 3.12 and 3.16.

3.D. Levels of differentiability

Throughout, U denotes an open subset of $\mathbb{R}^m$.

Definition 3.22. If a function $f : U \to \mathbb{R}^n$ is differentiable at every $a \in U$, then f is *differentiable on* U. Then the *derivative* of f, denoted df, is the function $U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ given by $a \mapsto df_a$.

3.D.1. Continuous differentiability

Recall from definition 1.26 that the set $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ of all linear maps $\mathbb{R}^m \to \mathbb{R}^n$ is a metric space under the operator norm. We use this metric space structure to make the following definition.

Definition 3.23. If $f : U \to \mathbb{R}^n$ is a differentiable function and the derivative $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ is continuous, then f is *continuously differentiable*.

Lemma 3.24. *A function $f : U \to \mathbb{R}^n$ is continuously differentiable if and only if the component functions $f_j : U \to \mathbb{R}$ are all continuously differentiable for $j = 1, \ldots, n$.*

Proof. We will use lemma 3.12, which tells us that f is differentiable if and only if f_j is differentiable for all $j = 1, \ldots, n$, and moreover that $df_{j,a} = \pi_j \circ df_a$ for all $a \in U$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "df_j"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \arrow[d, "\pi_j \circ -"] \\ & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \end{tikzcd}$$

You will verify in exercise 3.25(a) below that the map "postcompose with π_j" map $\pi_j \circ - : \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous. Thus, if f is continuously differentiable, then df_j is the composite of the two continuous maps df and $\pi_j \circ -$, so it is itself continuous.

Conversely, suppose that df_j is continuous for all j. Let us show that df is continuous at every $a \in U$. Suppose $\epsilon > 0$. Since df_j is continuous, there exists $\delta_j > 0$ such that

$$\|df_{j,a} - df_{j,b}\| < \frac{\epsilon}{\sqrt{n}}$$

for all $b \in U$ such that $|a - b| < \delta_j$. Let $\delta := \min\{\delta_1, \ldots, \delta_n\}$. Then $\|df_a - df_b\| < \epsilon$ for all $b \in U$ such that $|a - b| < \delta$, as you will verify in exercise 3.25(b) below. □

Exercise 3.25. (a) Check that the "postcompose with π_j" map $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ (ie, the one given by $\phi \mapsto \pi_j \circ \phi$) is continuous. Alternatively, prove the more general statement given in exercise 1.29 (which is not really much harder).

(b) With notation as in the latter part of the proof above, verify that $\|df_a - df_b\| < \epsilon$.

Theorem 3.26. *A function $f : U \to \mathbb{R}^n$ is continuously differentiable if and only if the partial derivatives $\partial_i f_j : U \to \mathbb{R}$ exist and are continuous for all $i = 1, \ldots, m$ and $j = 1, \ldots, n$.*

Proof. Thanks to lemma 3.24, we can assume without loss of generality that $n = 1$. If f is differentiable, then lemma 3.16 tells us that $\partial_i f = df(e_i)$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "\partial_i f"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \arrow[d, "\text{“evaluate at } e_i\text{”}"] \\ & \mathbb{R} \end{tikzcd}$$

The "evaluate at e_i" map is continuous by exercise 1.27. If f is continuously differentiable, then df is continuous, so we conclude that the composite $\partial_i f$ is also continuous.

The converse is the hard part of this proof. Recall from exercise 3.18 above that $\partial_i f$ existing for all i is *not* sufficient to guarantee differentiability; we will need to use the fact that $\partial_i f$ exists *and is continuous* to conclude differentiability.

□

Exercise 3.27. If you haven't alraedy done it, do exercise 1.27.

3.D.2. C^n and smooth functions

Definition 3.28. For any positive integer n, we say that a function $f : U \to \mathbb{R}^n$ is C^n if the partial derivatives $\partial_i f_j$ exist and are C^{n-1}.

Theorem 3.26 tells us that $f : U \to \mathbb{R}^n$ is C^1 if and only if it is continuously differentiable.

Definition 3.29. We say that a function $f : U \to \mathbb{R}^n$ is C^∞ or *smooth* if it is C^n for all positive integers n.

3.E. Inverse function theorem

4. Manifolds

4.A. Defining Manifolds

4.A.1. Charts and atlases

Throughout, let X be a metric space.

Definition 4.1. A *chart* on X is a pair (U, x) consisting of an open subset $U \subseteq X$ and a function $x : U \to \mathbb{R}^n$ which is a homeomorphism onto an open subset of $\mathbb{R}^n$ for some n. We call the functions $x_i := \pi_i \circ x : U \to \mathbb{R}$ *coordinate functions* on U with respect to the chart (U, x), and for any $P \in U$, the list of numbers $x_1(P), \ldots, x_n(P)$ are the *coordinates* of P with respect to the chart (U, x).

Exercise 4.2. Let $X = \mathbb{R}^2$ and let U denote $\mathbb{R}^2$ minus the non-negative real axis. Let $\text{polar} : U \to \mathbb{R}^2$ be the function which sends a point P to its polar representation $(r(P), \theta(P))$, where $r(P) = |P|$ and $\theta(P)$ is the angle in radians strictly between 0 and 2π formed between P and the non-negative real axis. Show that this function is a homeomorphism onto its image.

Definition 4.3. Suppose (U, x) and (V, y) are charts on X. The *transition map* from (U, x) to (V, y) is the function $y \circ x^{-1}$.

$$x(U \cap V) \xrightarrow{x^{-1}} U \cap V \xrightarrow{y} y(U \cap V)$$

Definition 4.4. The two charts (U, x) and (V, y) on X are *smoothly compatible*, or just *compatible*, if both of the transition maps between (U, x) and (U, y) are smooth.

Exercise 4.5. Let $\text{id} : \mathbb{R}^2 \to \mathbb{R}^2$ denote the identity map and $\text{polar} : U \to \mathbb{R}^2$ the polar representation map from exercise 4.2. Show that $(\mathbb{R}^2, \text{id})$ and (U, polar) are compatible charts.

Exercise 4.6. Show that the relation "is compatible with" is *not* an equivalence relation on the set of all charts on X. *Hint*. The function $\mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^3$ is a smooth homeomorphism, but its inverse is not differentiable.

Exercise 4.7. Suppose (U, x) and (V, y) are compatible charts, where $x : U \to \mathbb{R}^m$ and $y : V \to \mathbb{R}^n$, and that $U \cap V$ is nonempty. Show that $m = n$. *Hint*. Pick a point $P \in U \cap V$, and consider the derivatives of $x \circ y^{-1}$ at $y(P)$ and of $y \circ x^{-1}$ at $x(P)$.

Definition 4.8. An *atlas* on X is a collection $\mathcal{A}$ of charts such that covers X. In other words,

$$\bigcup_{(U,x)\in\mathcal{A}} U = X.$$

An atlas is *maximal* if there is no strictly larger atlas containing it.

Exercise 4.9. Prove that every atlas $\mathcal{A}$ is contained in a unique maximal atlas. *Hint*. Let $\mathcal{A}'$ be the set of all charts that are compatible with all of the charts in $\mathcal{A}$. Prove that $\mathcal{A}'$ is an atlas (caution: keep exercise 4.6 in mind). Then prove that it is maximal, and it is the only maximal atlas containing $\mathcal{A}$.

Exercise 4.10. Let $f : X \to Y$ be a continuous function and $\mathcal{A}$ an atlas on Y. Define

$$f^*\mathcal{A} = \{(f^{-1}(U))\}$$

4.A.2. Definition and examples

Definition 4.11. A *smooth manifold*, or just a *manifold*, is a pair $(X, \mathcal{A}_X)$ where X is a metric space and $\mathcal{A}_X$ is a maximal atlas. Often, write simply "X" in place of the pair $(X, \mathcal{A}_X)$, and sometimes $\mathcal{A}_X$ is called the "manifold structure on X."

Of course, exercise 4.9 tells us that any not-necessarily-maximal atlas on a metric space X defines a manifold structure on X in a canonical way. When constructing examples, it is usually useful to consider non-maximal atlases. But, when dealing with abstract manifolds, we will often find it convenient to assume that the atlas that the manifold comes equipped with is already maximal, which is the reason that the definition is made the way it is.

Example 4.12. The single chart $(\mathbb{R}^n, \mathrm{id})$, where $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$ is the identity map, defines an atlas on $\mathbb{R}^n$. We can extend this to a maximal atlas by exercise 4.9, called the *euclidean atlas* on $\mathbb{R}^n$. Unless explicitly specified otherwise, we always regard $\mathbb{R}^n$ as a manifold by equipping it with the euclidean atlas. By exercise 4.5, the polar representation chart (U, polar) is an element of the euclidean atlas.

Exercise 4.13. The single chart $(\mathbb{R}, f)$ where $f : \mathbb{R} \to \mathbb{R}$ is the function $f(x) = x^3$ is an atlas on $\mathbb{R}$. Show that the corresponding maximal atlas is *not* the euclidean atlas.

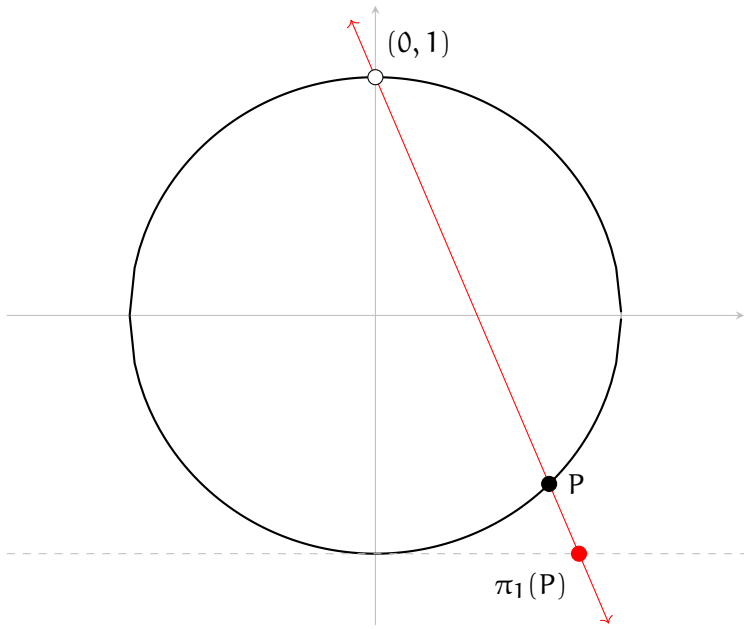

Figure 4.16.: Given a point P on $S^1 \setminus \{(0,1)\}$, we draw a line connecting P to $(0,1)$. The point where that line intersects the line $\mathbb{R} \times \{-1\}$ is the *projection of* P *from* $(0,1)$ *onto* $\mathbb{R} \times \{-1\}$, denoted $\pi_1(P)$.

Exercise 4.14. Let S^1 denote the set of points in $\mathbb{R}^2$ along the unit circle. In other words,

$$S^1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Given a point $P \in S^1 \setminus \{(0,1)\}$, the *projection of* P *from* $(0,1)$ *onto* $\mathbb{R} \times \{-1\}$, which we will denote $\pi_1(P)$, is defined as the point of intersection between the line $y = -1$ and the line passing through P and $(0,1)$. See figure 4.16. Similarly, for $P \in S^1 \setminus \{(0,-1)\}$, we can define the projection of P from $(0,-1)$ onto $\mathbb{R} \times \{1\}$, denoted $\pi_2(P)$. Show that π_1 and π_2 together define an atlas on S^1.

Exercise 4.15. Let S^n denote the set of points in $\mathbb{R}^{n+1}$ of norm 1. Generalize exercise 4.14 and show that "projection from the north pole" and "projection from the south pole" define an atlas on S^n.

Exercise 4.17 (Open submanifolds)**.** Let X be a manifold and $U \subseteq X$ an open subset. Show that

$$\mathcal{A}_U = \{(V,x) \in \mathcal{A}_X : V \subseteq U\}$$

is a maximal atlas on U. We call U, equipped with this maximal atlas, an *open submanifold* of X.

4.B. Smooth functions

4.B.1. Smooth real-valued functions

Definition 4.18. Let X be a manifold. A *smooth function* $f : X \to \mathbb{R}$ is a function such that, for any chart $(U, x) \in \mathcal{A}_X$, the function $f \circ x^{-1}$ is smooth.

$$\phi(U) \xrightarrow{x^{-1}} U \xrightarrow{f} \mathbb{R}$$

This definition might seem a bit annoying at first glance, since the maximal atlas $\mathcal{A}_X$ has *many, many* charts. The following exercise shows that we don't need to check all of the charts.

Exercise 4.19. Let X be a manifold and $\mathcal{A} \subseteq \mathcal{A}_X$ an atlas. If $f : X \to \mathbb{R}$ is a function such that $f \circ x^{-1}$ is smooth for any $(U, x) \in \mathcal{A}$, show that f is smooth.

Definition 4.20. Let X be a manifold and $(U, x) \in \mathcal{A}_X$ a chart. Recall that we then have coordinate functions $x_1, \ldots, x_n : U \to \mathbb{R}$, where $x_i = \pi_i \circ x$. Given a smooth function $f : X \to \mathbb{R}$ and a point $P \in U$, we define the *partial derivative of* f *at* P *with respect to* x_i by

$$\left.\frac{df}{dx_i}\right|_P = \partial_i(f \circ x^{-1})(x(P)).$$

In other words, df/dx_i is the function $\partial_i(f \circ x^{-1}) \circ x : U \to \mathbb{R}$.

4.B.2. Smooth functions between manifolds

Definition 4.21. Suppose X and Y are manifolds and $f : X \to Y$ is a function. Then f is *smooth* if, for every chart $(V, y) \in \mathcal{A}_Y$ and every chart $(U, x) \in \mathcal{A}_X$ such that $f(U) \subseteq V$, the

function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth.

$$\begin{tikzcd}
X \arrow[r, "f"] & Y \\
U \arrow[u, hook] \arrow[r, "f"] \arrow[d, "x"'] & V \arrow[u, hook] \arrow[d, "y"] \\
x(U) \arrow[r, "y \circ f \circ x^{-1}"] \arrow[d, hook] & y(V) \arrow[d, hook] \\
\mathbb{R}^m & \mathbb{R}^n
\end{tikzcd}$$

Once again, we don't need to check all possible charts. The precise statement is a bit complicated though; there's a long string of quantifiers.

Exercise 4.22. Suppose X and Y are manifolds and $f : X \to Y$ is a function. Suppose further that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_Y$ such that, for every $(V, y) \in \mathcal{A}$, there exists an atlas $\mathcal{A}' \subseteq \mathcal{A}_{f^{-1}(V)}$, such that for every $(U, x) \in \mathcal{A}'$, the function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth. Show that f is smooth.

Exercise 4.23. If $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}^n$ is a function, show that f is smooth in the sense of definition 3.29 if and only if it is smooth in the sense of definition 4.21.

Exercise 4.24. Show that a function $f : X \to \mathbb{R}$ is smooth in the sense of definition 4.18 if and only if it is smooth in the sense of definition 4.21.

4.B.3. Diffeomorphisms

Definition 4.25. Suppose X and Y are manifolds. A function $f : X \to Y$ is a *diffeomorphism* if it is smooth, bijective, and the inverse function $f^{-1} : Y \to X$ is also smooth. If there exists a diffeomorphism $f : X \to Y$, then X and Y are *diffeomorphic*.

Exercise 4.26. Regard $\mathbb{R}$ as a manifold in two different ways: once with the euclidean atlas, and next with the maximal atlas containing $x \mapsto x^3$. Show that these two manifolds are diffeomorphic.

Exercise 4.27. If X is a manifold and $(U, x) \in \mathcal{A}_X$, prove that $x : U \to x(U)$ is a diffeomorphism.

5. Tangent Spaces

5.A. Tangent space

5.A.1. Curves based at a point

Definition 5.1. Let X be a manifold and $P \in X$ a point. A *curve in* X *based at* P is a smooth function $\gamma : (-1, 1) \to X$ such that $\gamma(0) = P$.

Definition 5.2. Let X be a manifold and $P \in X$ a point. Two curves γ_1, γ_2 in X based at P are *equivalent* if, for all $(U, x) \in \mathcal{A}_X$ such that $P \in U$, we have

$$d(x^{-1} \circ \gamma_1)_0 = d(x^{-1} \circ \gamma_2)_0.$$

As usual, the first point of order is to check that we don't need to check all possible charts.

Exercise 5.3. Show that, if γ_1, γ_2 are two curves in X based at P and there exists a chart $(U, x) \in \mathcal{A}_X$ such that $P \in U$ and

$$d(x^{-1} \circ \gamma_1)_0 = d(x^{-1} \circ \gamma_2)_0,$$

then γ_1 is equivalent to γ_2.

Definition 5.4. For a manifold X and a point $P \in X$, we define the *tangent space of* X *at* P, denoted $T_P X$, to be the set of equivalence classes of curves in X based at P. Elements of $T_P X$ are called *tangent vectors* at P.

5.A.2. Vector space structure on the tangent space

5.A.3. Derivations

5.B. Pushing tangent vectors forward

Let $f : X \to Y$ be a smooth map of manifolds and $P \in X$ a point. Given a curve γ in X based at P, observe that $f \circ \gamma$ is a curve in Y based at $f(P)$.

Lemma 5.5. *If* $f : X \to Y$ *is a smooth map of manifolds,* $P \in X$*, and* γ_1 *and* γ_2 *are equivalent curves in* X *based at* P*, then* $f \circ \gamma_1$ *and* $f \circ \gamma_2$ *are equivalent curves in* Y *based at* $f(P)$*.*

This allows us to make the following definition, which in some sense is the "ultimate" generalization of derivatives.

Definition 5.6. Let $f : X \to Y$ be a smooth map of manifolds. Given a tangent vector $v \in T_P X$ defined by a curve γ, we define its *pushforward* $f_* v \in T_{f(P)} Y$ to be the tangent vector defined by $f \circ \gamma$. This defines a *pushforward map* $f_* : T_P X \to T_{f(P)} Y$.

Proposition 5.7. *Let* $f : X \to Y$ *be a smooth map of manifolds and* $P \in X$ *a point. Then* $f_* : T_P X \to T_{f(P)} Y$ *is a linear map.*

Bibliography